CultivaX

A Hybrid, Context-Aware Agricultural Intelligence and Service Coordination Platform

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Software Requirements Specification (SRS)

CultivaX: An Adaptive Agricultural Intelligence & Ecosystem Platform

Team Information

- **Project Title:** CultivaX
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- **Pilot Region:** Haryana (Pan-India Design)

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Software Requirements Specification (SRS) 1
Revision History 4
1. Introduction 5
1.1 Purpose 5
1.2 Intended Audience and Reading Suggestions 5
1.3 Product Scope 6
1.4 Definitions, Acronyms, and Abbreviations 6
1.5 References 6
2. Overall Description 7
2.1 Product Perspective 7
2.2 Product Functions 7
2.3 User Classes and Characteristics 8
2.4 Operating Environment 8
2.5 Design and Implementation Constraints 8
2.6 User Documentation 9
2.7 Assumptions and Dependencies 9
3. External Interface Requirements 10
3.1 User Interfaces 10
3.2 Communication Interfaces 11
3.3 Software Interfaces 12
3.4 Hardware Interfaces 12
3.5 Data Synchronization Interface 13
4. System Features 14
4.1 Crop Lifecycle Management 14
4.2 Context-Aware Crop Guidance 14
4.3 Stakeholder Discovery and Listings 15
4.4 Contextual Prioritization of Listings 15
4.5 Service Requests and Booking Flow 15
4.6 In-App Communication 16
4.7 Trust and Rating System 16
4.8 Policy and Scheme Awareness 17
4.9 Administrative Governance 17
4.10 Offline and Low-Connectivity Support 17
5. Other Non-Functional Requirements 18
5.1 Performance Requirements 18
5.2 Reliability Requirements 18
5.3 Availability Requirements 18
5.4 Security Requirements 18
5.5 Privacy and Data Ownership Requirements 19
5.6 Usability Requirements 19
5.7 Scalability Requirements 19

5.8 Legal and Ethical Requirements 19
5.9 Maintainability Requirements 20
5.10 Auditability and Logging Requirements 20
5.11 Intelligent System Requirements 20
6. Other Requirements 21
6.1 Data and Information Management Requirements 21
6.2 Localization and Regional Configuration 21
6.3 Language and Accessibility Requirements 21
6.4 Analytics and Insights 21
6.5 Compliance and Moderation 22
6.6 Future Enhancements (Out of Phase-1 Scope) 22
6.7 Explicit Exclusions 22
Appendix A – Glossary 23
Appendix B – Analysis Models 24
Appendix C – To Be Determined (TBD) List 25

Revision History

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1.0	28-01-2026	Initial SRS Draft	Team CultivaX

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements of **CultivaX**, an adaptive agricultural platform designed to support farmers and other agricultural stakeholders throughout the crop lifecycle.

The purpose of this document is to:

- clearly specify system behavior and constraints,
- establish a shared understanding among stakeholders,
- serve as a contractual reference for design, development, testing, and deployment.

This document focuses strictly on **what the system must do**, not **how it will be implemented**.

This SRS serves as the contractual baseline for all subsequent development phases.

1.2 Intended Audience and Reading Suggestions

This document is intended for:

- instructors and evaluators,
- developers and designers,
- testers and quality assurance teams,
- project managers,
- agricultural domain stakeholders.

Reading suggestions:

- Sections **1–2** provide system context and scope.
- Section **4** details functional behavior.
- Section **5** defines quality, reliability, and security requirements.

1.3 Product Scope

CultivaX is a **pan-India, one-stop agricultural ecosystem platform** that integrates crop lifecycle intelligence, service discovery, stakeholder coordination, and policy awareness.

The system is designed to:

- provide **context-aware, stage-based crop guidance**,
- enable discovery and coordination among multiple agricultural stakeholders,
- support **non-binding service requests** without enforced transactions,
- function reliably under **intermittent or limited internet connectivity**.

CultivaX follows a **trust-first philosophy**, prioritizing farmer control, transparency, and voluntary participation over enforcement or dependency.

CultivaX incorporates intelligent recommendation and prioritization mechanisms that may be implemented using data-driven models, open-source artificial intelligence systems, or external AI services. The system is designed to support real, continuously improving intelligence rather than static logic, while ensuring that all final decisions remain under user control.

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
Crop Timeline	Structured stages of a crop from sowing to post-harvest
Stakeholder	Any user role within the agricultural ecosystem
Hybrid Interface	UI combining dashboards/lists with optional map view
Non-binding Request	A request that does not constitute a confirmed transaction
Offline Mode	Limited access to previously synchronized data

1.5 References

1. IEEE Std 830-1984 – Software Requirements Specifications
2. Software Engineering Lab Manual – Faculty of Technology, University of Delhi

2. Overall Description

2.1 Product Perspective

CultivaX is a **standalone, web-based agricultural platform** designed to coordinate intelligence, services, and stakeholders within a single ecosystem.

The system adopts a **hybrid interaction model**:

- primary interaction through structured dashboards and lists,
- optional map view to support distance-aware decisions.

The platform is architected for **pan-India deployment**, with **Haryana used as a pilot region** for initial validation and demonstration.

CultivaX is explicitly designed to remain **usable under limited or unreliable internet conditions** through offline data access and deferred synchronization.

CultivaX functions as a coordination and intelligence platform and does not directly execute, supervise, or guarantee real-world agricultural operations. All actions resulting from system recommendations or interactions are voluntarily performed by users outside the system.

2.2 Product Functions

At a high level, CultivaX provides:

- Crop lifecycle tracking and guidance
- Contextual, stage-aware recommendations
- Discovery of services, workers, equipment, inputs, storage, logistics, and buyers
- Submission of non-binding service requests
- In-app communication with deferred message delivery when offline
- Multi-dimensional trust and rating mechanisms
- Access to government schemes and policy information
- Optional map-based visualization of listings
- Administrative moderation and analytics
- Limited offline access to previously synchronized data

2.3 User Classes and Characteristics

CultivaX supports the following user classes:

- **Farmer** – primary user; may experience intermittent connectivity
- **Agricultural Worker** – service provider with skill-based listings
- **Equipment Owner** – rental-based service provider
- **Service Provider** – specialized agricultural services
- **Input Seller** – chemical and organic inputs
- **Storage Provider** – warehouses and cold storage
- **Logistics Provider** – transport services
- **Agricultural Advisor** – expert guidance
- **Buyer / Aggregator** – post-harvest coordination
- **Administrator** – governance and system control

2.4 Operating Environment

- Web-based access via modern browsers
- Works on common consumer devices that can run modern web browsers.
- Internet connectivity required for real-time interactions
- **Offline support for previously synchronized data**
- English and Hindi supported by default
- Architecture supports additional Indian languages

2.5 Design and Implementation Constraints

- Payments excluded from Phase-1
- Offline mode limited to read-only advisory and saved data
- Messaging queued and delivered upon reconnection
- Role-based authentication and verification required
- Listings require periodic re-confirmation
- No enforcement of agricultural practices

2.6 User Documentation

- Role-specific onboarding guides
- Embedded help and prompts
- Administrator usage documentation

2.7 Assumptions and Dependencies

- Users may experience intermittent connectivity
- External data sources may be periodically unavailable
- Offline data may be temporarily outdated
- Sensor-based inputs are future extensions

3. External Interface Requirements

3.1 User Interfaces

CultivaX provides a **hybrid user interface** that combines structured dashboards with optional spatial visualization. The interface adapts based on user role and context while maintaining consistency across the platform.

3.1.1 General UI Characteristics

- Role-based dashboards for each stakeholder
- List- and card-based views as the default interaction mode
- Optional **Map View** available via toggle or tab
- Clear visual indicators for priority, status, and trust
- Multilingual support (English and Hindi by default)
- Responsive design for use on common consumer devices

3.1.2 Hybrid Discovery Interface

The system supports two complementary discovery modes:

A. Marketplace View (Default)

- Displays listings as ranked cards or lists
- Supports advanced filters, including:
    - distance
    - price
    - availability
    - rating / trust score
    - crop relevance
    - service type
    - region
- Accessible to all relevant stakeholders

B. Optional Map View

- Displays listings as pins on a map

- Each pin includes:
    - approximate location
    - service radius or coverage area
- Clicking a pin opens the corresponding listing card
- Map view complements, but does not replace, list-based discovery

3.1.3 Contextual Priority Indicators

Based on crop lifecycle stage and context:

- Relevant listings are visually highlighted
- Priority tags (e.g., “Recommended for Harvest Stage”) are displayed
- Non-relevant listings remain accessible but de-emphasized

This behavior applies consistently across list and map views.

3.1.4 Offline User Interface Behavior

When internet connectivity is unavailable or unstable:

- Previously synchronized data remains accessible in read-only mode
- Cached information includes:
    - crop timelines and last known recommendations
    - saved contacts and vendors
    - policy information and deadlines
- A visible indicator informs users that data may be outdated
- Actions requiring connectivity are disabled or deferred

3.2 Communication Interfaces

3.2.1 In-App Messaging

CultivaX provides in-app messaging between users with the following characteristics:

- Messages are exchanged only between authenticated users
- Contact details are shared only after mutual consent

- Users may optionally choose to make contact details public

3.2.2 Offline Messaging Behavior

- Messages composed while offline are **queued locally**
- Queued messages are sent automatically upon reconnection
- Users are informed of message delivery status
- Real-time chat is unavailable during offline periods

3.2.3 SMS Notification Interface

The system supports SMS notifications for critical events, including:

- important crop-stage alerts
- service request updates
- policy deadlines

SMS notifications are intended as a fallback mechanism and do not replace core platform functionality.

3.3 Software Interfaces

CultivaX may interface with external software systems, including:

- weather data providers
- government policy and scheme portals
- map and geolocation services
- messaging and notification services

These integrations are abstracted and configurable to support regional variability and future extensions.

3.4 Hardware Interfaces

The system does not require specialized hardware.
CultivaX is designed to operate on:

- standard consumer devices

- devices with intermittent connectivity
- browsers supporting modern web standards

Sensor-based inputs are not required in Phase-1 and are considered future enhancements.

3.5 Data Synchronization Interface

CultivaX supports deferred synchronization:

- Local data is updated when connectivity is restored
- Conflicts are resolved using server-side authority
- Offline edits are limited to message drafting and request preparation

4. System Features

4.1 Crop Lifecycle Management

4.1.1 Description

CultivaX shall enable farmers to manage crops as structured lifecycle instances, each progressing through defined stages from sowing to post-harvest.

4.1.2 Functional Requirements

- **FR-1:** The system shall allow a farmer to create a crop instance with basic details such as crop type, region, and start date.
- **FR-2:** The system shall maintain a predefined sequence of lifecycle stages for each crop.
- **FR-3:** The system shall display the current stage of a crop timeline to the farmer.
- **FR-4:** The system shall allow multiple crop instances to be managed simultaneously.

4.2 Context-Aware Crop Guidance

4.2.1 Description

CultivaX shall provide stage-aware recommendations aligned with the crop lifecycle and regional context.

4.2.2 Functional Requirements

- **FR-5:** The system shall generate recommendations based on the current crop lifecycle stage.
- **FR-6:** The system shall adapt recommendations when crop progress deviates from the expected timeline.
- **FR-7:** The system shall allow farmers to override or ignore recommendations.
- **FR-8:** The system shall record farmer overrides for future improvement and analysis.
- **FR-9:** The system shall display the rationale for recommendations in a concise manner.
- **FR-10:** The system shall improve recommendation accuracy by learning from user feedback and historical crop outcomes.

4.3 Stakeholder Discovery and Listings

4.3.1 Description

CultivaX shall support discovery of services, resources, and stakeholders relevant to farming activities.

4.3.2 Functional Requirements

- **FR-11:** The system shall allow stakeholders to create and manage listings based on their role.
- **FR-12:** The system shall associate listings with geographic location and service coverage area.
- **FR-13:** The system shall present listings in list-based views with advanced filtering options.
- **FR-14:** The system shall provide an optional map-based view for spatial understanding.
- **FR-15:** The system shall allow periodic re-confirmation of listings to ensure data freshness.

4.4 Contextual Prioritization of Listings

4.4.1 Description

CultivaX shall prioritize relevant listings based on crop lifecycle stage and farmer context without restricting access to other options.

4.4.2 Functional Requirements

- **FR-16:** The system shall highlight listings relevant to the current crop lifecycle stage.
- **FR-17:** The system shall de-emphasize non-relevant listings without hiding them.
- **FR-18:** The system shall apply prioritization consistently across list and map views.

4.5 Service Requests and Booking Flow

4.5.1 Description

CultivaX shall enable non-binding service requests between stakeholders.

4.5.2 Functional Requirements

- **FR-19:** The system shall allow farmers to send non-binding service requests.

- **FR-20:** The system shall display requests as “Pending Confirmation” to service providers.
- **FR-21:** The system shall allow service providers to accept or decline requests.
- **FR-22:** The system shall support exchange of booking details without enforcing payment.

4.6 In-App Communication

4.6.1 Description

CultivaX shall support secure communication between stakeholders.

4.6.2 Functional Requirements

- **FR-23:** The system shall provide in-app messaging between authenticated users.
- **FR-24:** The system shall queue messages composed offline and send them upon reconnection.
- **FR-25:** The system shall share contact details only after mutual consent.

4.7 Trust and Rating System

4.7.1 Description

CultivaX shall implement a multi-dimensional trust and rating mechanism.

4.7.2 Functional Requirements

- **FR-26:** The system shall allow users to rate interactions across multiple dimensions. (e.g., "reliability, communication quality, service completion rate").
- **FR-27:** The system shall compute a composite trust score from individual metrics.
- **FR-28:** The system shall display visual trust indicators and historical trends.

4.8 Policy and Scheme Awareness

4.8.1 Description

CultivaX shall provide access to relevant agricultural policies and schemes.

4.8.2 Functional Requirements

- **FR-29:** The system shall display region-specific policies and schemes.
- **FR-30:** The system shall notify users of important deadlines.
- **FR-31:** The system shall support redirection to official portals for applications.

4.9 Administrative Governance

4.9.1 Description

CultivaX shall provide administrative controls to ensure safe and compliant operation.

4.9.2 Functional Requirements

- **FR-32:** The system shall allow administrators to manage users and listings.
- **FR-33:** The system shall allow administrators to moderate content and resolve disputes.
- **FR-34:** The system shall allow administrators to suspend or ban users if required.
- **FR-35:** The system shall provide usage analytics and activity monitoring.

4.10 Offline and Low-Connectivity Support

4.10.1 Description

CultivaX shall remain usable under intermittent connectivity conditions.

4.10.2 Functional Requirements

- **FR-36:** The system shall provide offline access to previously synchronized advisory data.
- **FR-37:** The system shall display indicators when operating in offline mode.
- **FR-38:** The system shall synchronize cached data upon reconnection.

5. Other Non-Functional Requirements

5.1 Performance Requirements

- **NFR-1:** The system shall load primary dashboard views within acceptable response time under normal conditions.
- **NFR-2:** The system shall support concurrent usage by multiple stakeholders without functional degradation.
- **NFR-3:** Listing discovery shall return results within 2 seconds for queries with up to 1000 records.
- **NFR-4:** Offline data access shall not require network availability once data is synchronized.

5.2 Reliability Requirements

- **NFR-5:** The system shall preserve previously synchronized data during intermittent connectivity.
- **NFR-6:** The system shall queue user actions such as messages and requests when offline.
- **NFR-7:** The system shall synchronize queued actions automatically upon reconnection.
- **NFR-8:** The system shall prevent data corruption through periodic backups with defined retention policies.

5.3 Availability Requirements

- **NFR-9:** The system shall be available for use except during scheduled maintenance periods.
- **NFR-10:** Offline access to cached advisory data shall remain available even when the system is temporarily unreachable.
- **NFR-11:** Critical notifications may be delivered through alternative channels such as SMS notifications.

5.4 Security Requirements

- **NFR-12:** The system shall enforce role-based authentication for all users.
- **NFR-13:** Sensitive user data shall be accessible only to authorized parties.
- **NFR-14:** In-app communication shall be restricted to authenticated users.
- **NFR-15:** Administrative actions shall be logged for audit and safety purposes.

5.5 Privacy and Data Ownership Requirements

- **NFR-16:** Users shall retain ownership of the data they provide to the platform.
- **NFR-17:** The system may retain administrative logs strictly for moderation, dispute resolution, and safety.
- **NFR-18:** Aggregated and anonymized data may be used to improve platform functionality.
- **NFR-19:** Personal data shall not be shared without explicit user consent.

5.6 Usability Requirements

- **NFR-20:** Users with basic literacy shall complete core tasks.
- **NFR-21:** The system shall support multilingual interaction, with English and Hindi available by default.
- **NFR-22:** Visual indicators shall clearly communicate system states such as offline mode, pending requests, and confirmations through proper iconography or text labels.

5.7 Scalability Requirements

- **NFR-23:** The system architecture shall support addition of new regions through configuration changes only, without modifying core codebase.
- **NFR-24:** The system shall support growth in users, listings, and interactions across India.
- **NFR-25:** Additional stakeholder roles and services shall be incorporable without breaking existing functionality.

5.8 Legal and Ethical Requirements

- **NFR-26:** The system shall not guarantee crop yield or financial outcomes.
- **NFR-27:** The system shall not replace certified agricultural experts or official advisories.
- **NFR-28:** The system shall not enforce the use of specific agricultural inputs or practices.
- **NFR-29:** The platform shall act as a facilitator and not assume legal responsibility for disputes between users.

5.9 Maintainability Requirements

- **NFR-30:** The system shall allow administrators to update content, policies, and configurations without downtime.
- **NFR-31:** The system shall maintain logs to assist in debugging and future improvements.
- **NFR-32:** The system shall support incremental feature enhancements across development phases.

5.10 Auditability and Logging Requirements

- **NFR-33:** The system shall maintain audit logs for critical administrative, moderation, and policy-related actions.
- **NFR-34:** Audit logs shall be accessible only to authorized administrative roles and retained for safety, compliance, and dispute resolution purposes.

5.11 Intelligent System Requirements

- **NFR-35:** The system shall support integration with intelligent components capable of processing contextual and historical data.
- **NFR-36:** The system shall allow replacement or enhancement of intelligent components without impacting core platform functionality.

6. Other Requirements

6.1 Data and Information Management Requirements

- **OR-1:** The system shall maintain structured records for users, listings, crop instances, messages, and service requests.
- **OR-2:** The system shall support periodic re-confirmation of listings to ensure data relevance and accuracy.
- **OR-3:** Historical interaction data (e.g., requests, ratings) shall be retained for trust computation and analytics.
- **OR-4:** Cached data used for offline access shall be clearly distinguishable from real-time data.

6.2 Localization and Regional Configuration

- **OR-5:** The system shall support region-specific configuration of crop timelines, services, and policies.
- **OR-6:** The system shall allow administrators to configure regional data without affecting other regions.
- **OR-7:** Haryana shall be used as the initial pilot region while maintaining pan-India configurability.

6.3 Language and Accessibility Requirements

- **OR-8:** The system shall support English and Hindi interfaces from the initial release.
- **OR-9:** The system architecture shall allow addition of other Indian languages without restructuring.
- **OR-10:** Text, alerts, and labels shall be designed to remain understandable across different literacy levels.

6.4 Analytics and Insights

- **OR-11:** The system shall collect anonymized usage data for platform improvement.
- **OR-12:** Administrators shall have access to analytical summaries such as:
    - user activity trends,
    - listing distribution,
    - service demand patterns.

- **OR-13:** Analytics shall not expose personally identifiable user data.

6.5 Compliance and Moderation

- **OR-14:** The system shall provide mechanisms for users to report misuse or inappropriate behavior.
- **OR-15:** Administrators shall be able to review reports and take corrective action.
- **OR-16:** Administrative actions such as suspension or banning shall be logged for accountability.

6.6 Future Enhancements (Out of Phase-1 Scope)

The following capabilities are explicitly identified as **future enhancements** and are not required in Phase-1:

- integrated payment and transaction processing,
- automated booking confirmation,
- sensor-based farm data integration,
- advanced predictive analytics,
- real-time third-party system synchronization.

These features may be incorporated in later phases without altering core system assumptions.

6.7 Explicit Exclusions

CultivaX explicitly does **not**:

- guarantee crop yield or income outcomes,
- automate or control physical agricultural machinery,
- replace certified agricultural experts or official advisories,
- enforce use of specific fertilizers, chemicals, or services,
- act as a financial intermediary or payment processor,
- assume legal liability for disputes between stakeholders.

Appendix A – Glossary

Term	Definition
CultivaX	A trust-first agricultural ecosystem platform integrating crop intelligence and service coordination
Crop Instance	A specific occurrence of a crop managed by a farmer for a given season
Crop Timeline	A structured sequence of stages representing a crop's lifecycle
Stakeholder	Any user role participating in the CultivaX ecosystem
Service Radius	The geographic area served by a listing
Hybrid Interface	A UI model combining list-based interaction with optional map visualization
Non-binding Request	A service request that does not constitute a confirmed or paid transaction
Offline Mode	Limited access to previously synchronized data during connectivity loss
Trust Score	A composite rating derived from multiple evaluation dimensions

Appendix B – Analysis Models

This SRS does not mandate specific analysis models at this stage.
However, the following models may be developed in later phases:

- Use Case Diagrams
- Class Diagrams
- Sequence Diagrams
- Data Flow Diagrams

These models will be addressed during **Phase 2: System Design (Modeling)**.

Appendix C – To Be Determined (TBD) List

The following items are intentionally deferred and may be finalized in later phases:

TBD ID	Description
TBD-1	Exact visual design and UI layout
TBD-2	Final crop-stage configurations per region
TBD-3	External data source providers
TBD-4	Detailed analytics metrics
TBD-5	Payment workflow design (Phase-2)
TBD-6	Exact criteria for trust score computation